

RegDivergence-101: An LLM Benchmark for Cross-Jurisdiction Regulatory Contradiction Detection in Life Sciences

CHUCHU WU*, Carnegie Mellon University, USA

ZHIYIN ZHOU, Pratt Institute, USA

JINGZHUO HU, University of Pennsylvania, USA

LIANG YOU, University of Pittsburgh, USA

Pharmaceutical sponsors developing a drug for both the United States and the European Union must reconcile guidance issued independently by the FDA and the EMA. Where the two agencies require *substantively the same thing*, a sponsor can file once; where they *diverge*, a single trial design risks rejection in one region; where one agency is *silent* on a point the other regulates, the sponsor must infer obligations. Today this reconciliation is performed manually by regulatory-affairs experts. We introduce **cross-jurisdiction regulatory divergence detection**: given an FDA requirement and an EMA requirement on the same topic, classify their relationship as AGREE, DIVERGE, or SILENT. SILENT is inherently directional (SILENT_FDA vs. SILENT_EMA); we record direction per pair and report per-direction F1 alongside the collapsed label. We release **RegDivergence-101**, a 101-pair expert-grounded **pilot** evaluation benchmark (labels grounded in three peer-reviewed FDA/EMA comparison studies and primary FDA/EMA/ICH guidance text; dual-annotation inter-annotator $\kappa = 0.85$), and systematically characterise a four-method baseline hierarchy: lexical heuristic (0.511 macro-F1, 95% CI [0.411–0.605]), NLI cross-encoder (0.233), obligation-level Graph-RAG (0.663 [0.570–0.747]), and flat LLM judge / Claude Haiku (0.830 [0.747–0.908]). Three directional observations emerge at pilot scale ($n = 101$): SILENT is semantically detectable but invisible to entailment-only formulations; pair-level obligation graphs improve over lexical methods but trail flat-LLM context (CIs partially overlapping); and corpus-level graph construction is the indicated architectural target for large-scale silent-detection. RegDivergence-101 is a pilot release establishing the task formulation and baseline hierarchy; four unrepresented regulatory domains and an expansion roadmap are described in §7.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; • **Information systems** → *Document filtering*; • **Applied computing** → *Health informatics*.

Additional Key Words and Phrases: large language models, LLM benchmark, regulatory NLP, AI for compliance, FDA/EMA divergence, contradiction detection, Graph-RAG, retrieval-augmented generation, generative AI, life sciences, AGREE/DIVERGE/SILENT classification

*Corresponding author.

Authors' Contact Information: Chuchu Wu, chuchuw@alumni.cmu.edu, Carnegie Mellon University, Pittsburgh, PA, USA; Zhiyin Zhou, zzhou20@pratt.edu, Pratt Institute, New York, NY, USA; Jingzhuo Hu, jingzhuoh00@gmail.com, University of Pennsylvania, Philadelphia, PA, USA; Liang You, liy121@pitt.edu, University of Pittsburgh, Pittsburgh, PA, USA.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

Manuscript submitted to ACM

ACM Reference Format:

Chuchu Wu, Zhiyin Zhou, Jingzhuo Hu, and Liang You. 2026. RegDivergence-101: An LLM Benchmark for Cross-Jurisdiction Regulatory Contradiction Detection in Life Sciences. In *Proceedings of 2026 2nd International Conference on Artificial Intelligence and Foundation Model (AIFM '26)*. ACM, New York, NY, USA, 12 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Bringing a medicine to market in both the US and EU requires satisfying two regulators that publish guidance separately and frequently *differently*. Gene-therapy sponsors face a 15-year long-term follow-up recommendation from FDA while EMA defers the duration to case-by-case risk assessment—same topic, materially different specificity. Analogous divergences span pediatric extrapolation, endpoint definitions in ulcerative colitis, biosimilar comparability, and estimand implementation. A sponsor who misses one divergence during protocol design can lose a year or more re-running a study.

Reconciling the two corpora is currently a manual, expert task. Yet the NLP building blocks for automating it all exist—document-level NLI (ContractNLI [1], DocNLI [2]), legal contradiction detection (LegalWiz [3], CLAUSE [4]), and Graph-RAG (Microsoft GraphRAG [5], GraphCompliance [6], RAGulating Compliance [7]). **No prior work combines them to detect divergence between two parallel regulatory corpora from different jurisdictions.** All existing contradiction work is *intra-document* (within one contract) or *single-jurisdiction* (contract-vs-statute).

This paper makes three contributions:

- (1) **Task formulation.** Cross-jurisdiction regulatory divergence detection as a three-way classification (AGREE / DIVERGE / SILENT) over pre-aligned FDA/EMA requirement pairs, grounded in published expert comparisons. The task is explicitly scoped to the *classification* stage given pre-aligned pairs; topic alignment is a non-trivial upstream problem addressed in §3.1 and §3.2. SILENT is treated as directional (SILENT_FDA vs. SILENT_EMA); we report per-direction F1 in Table 2.
- (2) **Pilot benchmark. RegDivergence-101**, a pilot 101-pair benchmark (59 from peer-reviewed FDA/EMA comparison studies, 42 derived from primary FDA/EMA/ICH guidance) spanning 13 topic areas, with a documented two-track pair-construction protocol, inter-annotator agreement ($\kappa = 0.85$), and bootstrap confidence intervals. Four regulatory domains are currently unrepresented; §7 describes the expansion roadmap.
- (3) **Baseline hierarchy with three directional observations.** Four methods systematically compared at pilot scale ($n = 101$), yielding directional empirical observations on the task’s structure and the path to a corpus-level solution. Observations are reported with bootstrap 95% CIs; confirmatory significance testing awaits a larger benchmark.

2 Related Work

Document-level NLI. ContractNLI [1] introduced document-level entailment/contradiction/not-mentioned classification with evidence extraction, and showed that contradiction detection lags entailment, especially under “negation by exception.” DocNLI [2] extended NLI to full documents.

Legal contradiction detection (2025–2026). LegalWiz [3] uses a multi-agent generation framework for legal contradiction detection but is confined to single-document regimes. Better Call CLAUSE [4] benchmarks 7,500+ perturbed contracts across ten anomaly categories and finds that LLMs miss subtle errors and struggle to *justify* them legally. Contradiction Detection in RAG Systems [8] uses LLMs as context validators. Two recent RAG-faithfulness studies bear directly on our LLM-judge and graph-RAG results: Chen et al. [9] probe whether a RAG system can recognise when retrieved context is incorrect, and Qian et al. [10] show that the surface relevance of retrieved evidence does not guarantee it *warrants* a claim—the same failure mode documented in our Observation 2, where modal-register similarity (RECOMMENDED vs. MANDATORY) misleads the pair-level graph.

Graph-RAG and regulatory compliance. Microsoft GraphRAG [5] builds entity graphs with community summaries for local and global queries. GraphCompliance [6] aligns *policy graphs* (regulatory requirements) with *context graphs* (organisational facts) for GDPR; it explicitly lists “handling conflicting regulatory interpretations” as open. RAGulating Compliance [7] builds an ontology-free regulatory knowledge graph for traceable QA.

Cross-jurisdiction regulatory benchmarks. The closest concurrent work is **Sino-US-DrugQA** [11] (11,871 MCQA items, US–CN), which shares our concept-alignment thesis but cannot express the SILENT class. Comparative-regulatory-science surveys document FDA/EMA divergence across dosing [12], oncology labelling [13], and multi-country SmPCs [14], but as human-authored comparisons, not ML benchmarks. RegDivergence-101 is the first FDA/EMA benchmark with an explicit SILENT class.

Foundation models and AI governance beyond text. The momentum of foundation models now spans multimodal scientific modelling [15], vector-graphics generation [16] and animation [17], and vision-language navigation [18], while parameter-efficient domain adaptation and ensembling of small language models [19] lower deployment cost. In high-stakes sectors such as healthcare, deep-learning deployment is increasingly governed by policy-aligned evaluation frameworks that weigh safety and regulatory constraints over raw accuracy [20]. RegDivergence-101 extends this momentum to cross-jurisdiction regulatory reasoning, a high-stakes text domain that remains underrepresented in current foundation-model evaluation.

Gap. Contradiction detection is intra-document or single-jurisdiction; Graph-RAG targets single-corpus QA or single-jurisdiction compliance; FDA/EMA divergence is documented manually but never automated. Our task sits precisely in this gap.

3 Task and Dataset

3.1 Task Definition

For a topic t , let f_t be an FDA requirement statement and e_t the corresponding EMA statement. The label y_t is:

- **AGREE** — both jurisdictions require substantively the same thing (same modal strength, same threshold, same scope of applicability). Paraphrase equivalence and synonymous modals (*shall* / *must*) are treated as AGREE; minor drafting differences that do not alter sponsor obligations are disregarded.
- **DIVERGE** — the requirements directly conflict: they impose different thresholds, different modal strengths (*shall* vs. *should*), different timelines, or mutually incompatible design specifications.

- **SILENT** — one jurisdiction does not address what the other regulates. SILENT is *directional* by construction:
 - **SILENT_FDA**: FDA is silent on this topic; an EMA requirement exists but no corresponding FDA requirement was found in the full published FDA corpus (guidance documents, Q&As, MaPPs).
 - **SILENT_EMA**: EMA is silent on this topic; an FDA requirement exists but no corresponding EMA requirement was found in the full published EMA corpus (guidelines, Q&As, reflection papers).

RegDivergence-101 records direction in the `silent.direction` field and reports per-direction F1 in Table 2. A unified **SILENT** label ($= \text{SILENT_FDA} \cup \text{SILENT_EMA}$) is retained in Table 1 for backward compatibility.

Epistemic status of SILENT labels. A SILENT label is an absence claim: it asserts that no counterpart requirement was found after an exhaustive corpus search. This carries irreducible epistemic uncertainty. We bound this uncertainty through (a) a documented corpus-search protocol (§3.3) and (b) a per-pair `doc-confidence` flag (HIGH / MODERATE). Downstream work should treat MODERATE-confidence SILENT pairs as soft labels.

Task scope and assumptions. This formulation assumes requirement pairs are *pre-aligned*—both the FDA and EMA sides address the same regulatory topic. Topic alignment is a non-trivial upstream problem: FDA and EMA guidance frequently address overlapping domains at inconsistent levels of specificity, making it unclear whether two requirements are “on the same topic” at all. RegDivergence-101 is constructed from expert-panel comparisons that *already provide aligned pairs* and from a two-track alignment procedure described in §3.2; we treat topic alignment as a separate, upstream task and refer readers to §7 for the corpus-level architecture that would encompass it.

3.2 Pair Construction Protocol

RegDivergence-101 is constructed via two tracks that differ in how topical correspondence between FDA and EMA requirements is established. Per-pair source-track attribution (`track`: A or B) is released with the dataset.

Track A — expert-comparison-derived pairs (59 pairs, 58.4%). All 59 pairs were extracted from three open-access peer-reviewed studies whose explicit purpose was comparative mapping of FDA and EMA requirements in a defined therapeutic domain:

- (1) Vieujean et al. [21] — ulcerative colitis endpoints and trial design (40 pairs).
- (2) Boesen et al. [22] — psychiatric drug-trial guidelines (10 pairs).
- (3) Schwarz and Decristoforo [23] — radiopharmaceutical nonclinical requirements (9 pairs).

In each source paper, domain experts had already established co-topicality through structured literature mapping or side-by-side dossier comparison. Our annotation task for Track A was to classify the *relationship* (AGREE / DIVERGE / SILENT) between requirements that the source authors had already deemed co-topical. We inherit, document, and cite their alignment judgements; per-pair source-study attribution is released with the dataset.

Track B — primary guidance pairs (42 pairs, 41.6%). For pairs derived from ICH E9(R1), E11A [24], and FDA/EMA biosimilar and gene-therapy guidance, topical alignment was established in two steps:

- (1) *Anchor identification.* Requirements were associated with ICH Common Technical Document (CTD) module and sub-topic designations (e.g., Module 5.3.5.1 for comparative bioavailability study reports). Requirements sharing the same CTD anchor were candidate pairs.
- (2) *Expert confirmation.* The primary annotator independently reviewed each candidate pair and confirmed or rejected co-topicality before the pair entered the dataset.

Specificity exclusion criterion. A candidate Track B pair was excluded if the FDA and EMA requirements operated at different levels of the regulatory hierarchy (e.g., one requirement addressing trial design at the *statistical analysis plan* level while the other addressed the same concept at the level of *general statistical principles*). Such pairs would conflate regulatory level-of-detail differences with substantive DIVERGE labels. A small number of candidate pairs were excluded on this basis (the count and representative cases are documented in the released dataset’s `excluded_pairs.jsonl` file). The complete exclusion-criteria checklist is released with the dataset so third parties can audit or extend our alignment judgements.

Each pair records `{id, topic, track, source, fda_text, ema_text, label, silent_direction, doc_confidence, disputed, annotation_notes}`. The combined release contains **101 pairs**: 42 AGREE, 38 DIVERGE, 21 SILENT, across 13 topic areas. We report macro-F1 to weight all classes equally.

3.3 Annotation Protocol

Annotators. Annotator A (C. Wu) is a solution architect for a regulatory-document AI platform; Annotator B (Z. Zhou) is a product manager for an AI-powered clinical-trial management platform. Neither holds formal RA credentialing; Track A expert-panel papers serve as the external human-expert anchor.

Stages. Stage 1: Wu assigned labels using a definitions-based framing (Track A labels inferred from source authors’ stated assessments; Track B labels assigned against full guidance text). Stage 2: Zhou independently re-labelled all 101 pairs under a sponsor-obligation framing to probe framing-independent stability. Both annotators are co-authors; the κ is a cross-framing consistency measure, not identical-instruction IAA. An independent third-annotator pass is planned for RegDivergence-500.

IAA. Human cross-framing $\kappa = 0.85$ (3-class, 101 pairs; observed agreement 0.90)—*substantial* per Landis and Koch [25], on par with MedNLI (0.65) and ContractNLI (≈ 0.72). SILENT is near-perfect ($\kappa = 0.98$); AGREE/DIVERGE more variable (≈ 0.79 – 0.81). A secondary LLM prompt-robustness check on a 36-pair sample yielded $\kappa = 0.542$ (LLM test-retest, not human IAA).

Disagreements (n = 10). Six resolved by source-paper language; four genuinely borderline pairs retain the Stage 1 label and are flagged `disputed = True` in the dataset with written justifications.

SILENT confidence. All 21 SILENT pairs carry `doc_confidence = MODERATE`: exhaustive corpus search was performed (full FDA and EMA guidance, Q&As, reflection papers) but absence claims remain epistemically soft. The HIGH/MODERATE audit is deferred to RegDivergence-500.

4 Baselines

4.1 Evaluation Context

RegDivergence-101 is an **evaluation benchmark**, not a training dataset. All four baselines are zero-shot or few-shot. Adjacent methods are separated by $\Delta = 0.15\text{--}0.60$ macro-F1 (min: Graph-RAG vs. LLM, $\Delta = 0.167$; CIs partially overlapping); bootstrap CI half-widths $\approx \pm 0.08\text{--}0.10$ at $n = 101$. Results are reported as a ranking trend, not pairwise significance.

4.2 Shared Evaluation Protocol

We evaluate four methods under a shared reporting framework: **stratified 5-fold splits** averaged over 5 random seeds, with **bootstrap 95% CI** (1,000 resamples). The protocol differs by method class. The lexical heuristic and NLI cross-encoder have tunable thresholds and use **nested** CV (inner folds tune thresholds, outer folds report). The Graph-RAG classifier and LLM judge have no tunable hyperparameters: the Graph-RAG classifier is reported over the same outer folds, and the LLM judge is scored directly on all 101 pairs. CIs for both LLM methods are bootstrap-only.

4.3 Lexical Heuristic

No neural model, no API—fully offline and deterministic. Combines TF-IDF cosine similarity with five conflict signals: negation asymmetry, divergence cue pairs (e.g., *single-arm* vs. *randomised*), modal-strength asymmetry (*must/shall* vs. *may/encouraged*), numeric-threshold mismatch, and one-sided specificity (concrete number vs. case-by-case deferral). A shared ICH citation is a strong AGREE signal.

4.4 NLI Cross-Encoder

`typeform/distilbert-base-uncased-mnli`, run in both directions; labels derived from contradiction/neutral/entailment probabilities with CV-tuned thresholds. We use DistilBERT-MNLI as a representative off-the-shelf NLI system. The structural SILENT gap observed in §6 (Observation 1) is model-agnostic: any NLI model lacking an explicit SILENT output class will map absence-of-regulation to *neutral*, collapsing SILENT recall. Substituting DeBERTa-v3-large or a modern cross-encoder would improve AGREE/DIVERGE F1 but would not resolve the SILENT structural gap without task-specific supervision.

4.5 Graph-RAG Pair-Level Classifier

For each text, Claude Haiku extracts a policy triplet $\langle \textit{subject}, \textit{obligation_level}, \textit{requirement}, \textit{conditions} \rangle$ where $\textit{obligation_level} \in \{\text{MANDATORY}, \text{PROHIBITED}, \text{RECOMMENDED}, \text{PERMITTED}, \text{SILENT}\}$. Only $\text{MANDATORY} \leftrightarrow \text{PROHIBITED}$ is a hard-coded DIVERGE rule; all other pairs are escalated to a second LLM call that reasons over the *obligation structure* of the aligned node pair, with explicit calibration that modal-register differences (RECOMMENDED vs. MANDATORY) do not automatically imply DIVERGE.

Within-model scope. Both the Graph-RAG classifier and the flat LLM judge use `claude-haiku-4-5-20251001`. The comparison between them is therefore a *within-model ablation*—with vs. without explicit

Table 1. Macro-F1, accuracy, and per-class F1 with bootstrap 95% confidence intervals for the four baselines ($n = 101$ pairs).

Method	Macro-F1 [95% CI]	Acc	AGREE F1 [CI]	DIVERGE F1 [CI]	SILENT F1 [CI]
Lexical heuristic ¹	0.511 [0.411–0.605]	0.541	0.718 [0.605–0.809]	0.466 [0.320–0.593]	0.334 [0.133–0.537]
NLI cross-encoder ²	0.233 [0.152–0.303]	0.238	0.279 [0.103–0.449]	0.383 [0.247–0.506]	0.025 [0.000–0.086]
Graph-RAG pair-level ³	0.663 [0.570–0.747]	0.673	0.754 [0.646–0.851]	0.586 [0.433–0.722]	0.640 [0.478–0.771]
LLM judge (Haiku)⁴	0.830 [0.747–0.908]	0.842	0.905 [0.831–0.965]	0.805 [0.696–0.892]	0.778 [0.609–0.914]

¹Nested 5-fold CV, 5 seeds; thresholds tuned on inner folds only. ²Same CV protocol, thresholds tuned on inner folds. ³No tunable hyperparameters; bootstrap CI only. ⁴claude-haiku-4-5-20251001, default temperature; bootstrap CI only. Estimated cost: ~\$0.10 for all 101 pairs.

obligation-graph structure—not an architectural comparison across model families. Conclusions about the relative value of graph structure should be confirmed with a second model family before generalisation.

4.6 LLM Judge (Flat Pairwise)

Claude Haiku with explicit AGREE / DIVERGE / SILENT definitions, prompted once per pair with both texts. No graph, no retrieval. Full prompt and code at Appendix A.

5 Results

Table 1 reports macro-F1, accuracy, and per-class F1 with bootstrap 95% confidence intervals for all four methods.

Table 2. Per-direction SILENT F1 for all four methods (bootstrap 95% CI, $n = 101$ pairs). SILENT.FDA ($n = 13$): FDA is silent, EMA regulates. SILENT.EMA ($n = 8$): EMA is silent, FDA regulates.

Method	SILENT.FDA F1 [95% CI]	SILENT.EMA F1 [95% CI]
Lexical heuristic	0.277 [0.000–0.545]	0.393 [0.000–0.706]
NLI cross-encoder	0.049 [0.000–0.167]	0.000 [0.000–0.000]
Graph-RAG pair-level	0.716 [0.516–0.882]	0.535 [0.267–0.762]
LLM judge (Haiku)	0.795 [0.588–0.952]	0.738 [0.429–0.952]

SILENT-FDA: 13 pairs (FDA silent, EMA active). SILENT-EMA: 8 pairs (EMA silent, FDA active). Wide CIs on SILENT-EMA reflect the small stratum ($n = 8$). Bootstrap 1,000 resamples; stratified by 3-class label.

Point estimates increase monotonically from lexical to Graph-RAG to the LLM judge. At $n = 101$ the marginal bootstrap CIs partially overlap, so we report these as a consistent ranking trend rather than a claim of statistical significance; a paired significance test is left to future work. **Lexical ceiling.** Characteristic errors share the signature of *high lexical overlap, opposed regulatory stance*: FDA “modified Mayo score of 5–9” vs. EMA “full Mayo score of 9–12” (same vocabulary, different instrument); FDA 15-year LTF vs. EMA case-by-case (same topic, different specificity). Resolving these requires representing what each requirement *asserts*, not surface similarity.

6 Analysis

6.1 What the Baseline Hierarchy Reveals

Observation 1 — SILENT is semantically detectable but invisible to NLI framing. SILENT $F1 = 0.025$ under NLI (no native SILENT class) vs. 0.778 under the LLM judge (explicit definition). This contrast is large and consistent across bootstrap resamples; it reflects a structural constraint—the SILENT class requires a formulation that can express absence, which NLI entailment cannot—rather than a model-capacity effect. We note this observation is based on a distilbert-class model; a higher-capacity NLI encoder would improve AGREE/DIVERGE $F1$ but would not resolve the absence-detection gap without task-specific supervision (§6).

Observation 2 — For Claude Haiku on this dataset, pair-level obligation graphs improve over lexical (+0.15 $F1$) but trail flat prompting (−0.17 $F1$); CIs partially overlap at pilot scale. The graph makes the obligation structure explicit (MANDATORY / RECOMMENDED / etc.), which helps distinguish clear stance differences from similar-sounding text. But it discards full-text context that flat prompting uses to separate *same-requirement* / *different-modal-register* (AGREE) from *same-topic* / *different-threshold* (DIVERGE). Pairs like gene-01 (FDA 15 years vs. EMA case-by-case) both extract as RECOMMENDED and the graph calls them AGREE; the flat LLM correctly identifies the threshold conflict from the text [10]. *Scope note.* Both the graph classifier and the flat LLM judge use Claude Haiku; this comparison is a within-model ablation (with vs. without explicit graph structure) rather than a general claim about graph methods vs. LLM methods. Replication with a second model family is left to future work.

Observation 3 — Two distinct problem scopes suggest two distinct architectural targets. The *pair-level scope* — given pre-aligned sentences, label AGREE/DIVERGE/SILENT — is largely addressed by the flat LLM judge (0.830), approaching a task ceiling. The *corpus-level scope* — “is EMA silent on this topic *anywhere* across its guideline corpus?” — requires finding aligned pairs, not just classifying them; a pairwise LLM cannot address this, but a corpus-level policy graph can. These are different tasks. This paper establishes baselines for the pair-level scope; §7 motivates the corpus-level architecture as the next contribution. This observation is architectural and directional; it is not a validated finding.

6.2 Source-Stratified Performance Analysis

To assess whether performance reflects source-study-specific patterns rather than genuine task competence, we analyse performance stratified by track and by source concentration.

Track-level. On AGREE/DIVERGE classes, the LLM judge scores *lower* on Track A ($0.769/0.745$) than Track B ($0.966/0.923$): exposure to source studies does not inflate performance. The macro- $F1$ gap (Track B 0.630 vs. Track A 0.765) is explained by SILENT support ($n = 0$ vs. $n = 21$), not memorisation.

Source-concentration. 40/101 pairs come from one source (Vieujean et al.). LLM judge on the JCC stratum ($n = 40$): 0.804 ; on all other pairs ($n = 61$): 0.844 . Performance is marginally *higher* outside JCC, ruling out domain-concentration inflation.

7 Future Work

Corpus-level policy graph. The correct long-term target is corpus-level silent detection and divergence discovery: (1) build per-jurisdiction policy graphs indexing all FDA/EMA guidance at corpus scale; (2) align requirement nodes via entity normalisation and embedding similarity; (3) classify aligned pairs as AGREE/DIVERGE/SILENT by obligation-structure comparison; (4) generate natural-language explanations. This architecture subsumes the pair-alignment assumption: alignment becomes a graph-construction step, not a pre-processing input.

RegDivergence-500. Four domains are unrepresented in the pilot: CMC, adaptive trial design/estimands, statistical analysis plan requirements, and labelling/SmPC. RegDivergence-500 (≈ 500 pairs) will cover all four via a community-annotation pipeline seeded from a ~ 45 -source FDA/EMA comparison inventory. Annotation guidelines are released with this paper. A third jurisdiction (PMDA) via FDA/EMA/PMDA concordance studies [12, 13] will add an ≈ 40 -pair FDA $\times$ PMDA subset. Multi-model replication and an independent third-annotator validation pass are planned. A cross-pair generalisation probe against Sino-US-DrugQA [11] will be the first test of whether conclusions transfer beyond the FDA/EMA pair.

8 Limitations

Topic alignment. Track B co-topicality judgements are single-annotator; Track A inherits source-study alignment methodology. Reproducibility depends on the released exclusion-criteria checklist (`data/alignment_protocol.md`).

Scale and coverage. 101 pairs from three source studies limits statistical power (directional observations, not confirmed findings). Oncology, rare diseases, medical devices, and vaccines are absent. RegDivergence-500 (§7) is required before domain-general conclusions.

SILENT labels. All 21 SILENT pairs carry `doc_confidence = MODERATE`; treat them as soft labels. The pair-level SILENT task is a proxy for the correct corpus-level formulation (§7).

Single-model evaluation. All LLM-dependent components (triplet extraction, graph classification, flat judge, annotation check) use Claude Haiku. The Graph-RAG vs. LLM comparison is a within-model ablation, not a cross-model comparison. The $\kappa = 0.542$ LLM consistency check reflects the same model’s label tendencies. Multi-model replication is deferred to RegDivergence-500.

Contamination. Haiku may have seen FDA/EMA guidance during pretraining. The source-stratified analysis (§6.2) provides evidence against inflation; future work should evaluate on post-cutoff guidance.

Use norms. RegDivergence-101 is a pilot benchmark: not sufficient for deployment decisions, not domain-generalizable without held-out evaluation, and not a sole criterion for AI procurement in regulatory-affairs workflows. Conclusions are specific to the FDA/EMA pair until cross-pair generalisation probes are run.

9 Conclusion

We introduced cross-jurisdiction regulatory divergence detection, released **RegDivergence-101**, a 101-pair expert-grounded **pilot** evaluation benchmark ($\kappa = 0.85$, documented two-track pair-construction protocol, per-direction SILENT annotation), and characterised a four-method baseline hierarchy with bootstrap CIs.

Three directional observations at pilot scale ($n = 101$, CIs partially overlapping for adjacent methods): SILENT requires explicit absence-aware formulation (NLI F1 0.03 $\rightarrow$ LLM F1 0.78); for Claude Haiku on this dataset, pair-level graph structure improves over lexical heuristics (+0.15 F1) but trails flat prompting (−0.17 F1); and the two-scope distinction (pair-level vs. corpus-level) motivates the corpus-level graph architecture as the next contribution. The flat LLM judge (0.830) establishes a pair-level performance reference at pilot scale; the corpus-level architecture proposed in §7 is the correct target for large-scale silence discovery. The benchmark is a pilot: four regulatory domains (CMC, clinical trial design, statistical analysis, labelling) remain unrepresented, and the planned RegDivergence-500 extension with expanded dual-annotation will establish the task at production scale.

Acknowledgments

This work was conducted independently. Benchmark pairs were derived from publicly available FDA/EMA guidance documents and open-access peer-reviewed comparison studies. LLM API calls used the Anthropic API (Claude Haiku, `claude-haiku-4-5-20251001`).

References

- [1] Yuta Koreeda and Christopher D. Manning. 2021. ContractNLI: A Dataset for Document-level Natural Language Inference for Contracts. In *Findings of the Association for Computational Linguistics: EMNLP 2021*. Association for Computational Linguistics, 1139–1150.
- [2] Wenpeng Yin, Dragomir Radev, and Caiming Xiong. 2021. DocNLI: A Large-scale Dataset for Document-level Natural Language Inference. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*. Association for Computational Linguistics, 4913–4922.
- [3] Ananya Mantravadi, Shivali Dalmia, Abhishek Mukherji, Nand Dave, Anudha Mittal, and Olga Pospelova. 2025. Legal-Wiz: A Multi-Agent Generation Framework for Contradiction Detection in Legal Documents. In *NeurIPS 2025 Workshop on Generative and Protective AI for Content Creation*. arXiv:2510.03418.
- [4] Manan Roy Choudhury, Adithya Chandramouli, Mannan Anand, and Vivek Gupta. 2026. Better Call CLAUSE: A Discrepancy Benchmark for Auditing LLMs’ Legal Reasoning Capabilities. In *Findings of the Association for Computational Linguistics: EACL 2026*. Association for Computational Linguistics, 5776–5818. arXiv:2511.00340.
- [5] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. 2024. From Local to Global: A GraphRAG Approach to Query-Focused Summarization. arXiv:2404.16130. arXiv:2404.16130 [cs.CL]
- [6] Jiseong Chung, Ronny Ko, Wonchul Yoo, Makoto Onizuka, Sungmok Kim, Tae-Wan Kim, and Won-Yong Shin. 2025. GraphCompliance: Aligning Policy and Context Graphs for LLM-Based Regulatory Compliance. arXiv:2510.26309. arXiv:2510.26309 [cs.AI]
- [7] Bhavik Agarwal, Hemant Sunil Jomraj, Simone Kaplunov, Jack Krolick, and Viktoria Rojkova. 2025. RAGulating Compliance: A Multi-Agent Knowledge Graph for Regulatory QA. arXiv:2508.09893. arXiv:2508.09893 [cs.CL]
- [8] Vignesh Gokul, Srikanth Tenneti, and Alwarappan Nakkiran. 2025. Contradiction Detection in RAG Systems: Evaluating LLMs as Context Validators for Improved Information Consistency. arXiv:2504.00180. arXiv:2504.00180 [cs.CL]
- [9] Yihang Chen, Pin Qian, Su Wang, Sipeng Zhang, Huan Xu, Shuhuai Lin, and Xinpeng Wei. 2026. Does RAG Know When Retrieval Is Wrong? arXiv:2605.14473. arXiv:2605.14473 [cs.CL]
- [10] Pin Qian, Su Wang, Xiaoyuan Wang, Yihang Chen, Wenxuan Xu, Qiaolin Yu, Shuhuai Lin, Sipeng Zhang, Junxian You, and Xinpeng Wei. 2026. Relevant Is Not Warranted: Evidence-Force Calibration for Cited RAG. arXiv:2605.28044. arXiv:2605.28044 [cs.CL]
- [11] Sino-US-DrugQA authors. 2026. Sino-US-DrugQA: A Benchmark for Evaluating Large Language Models in Cross-Jurisdictional Pharmaceutical Regulation. medRxiv preprint; <https://github.com/DodgeLU/Sino-US-DrugQA>. Preprint (medRxiv, 17 Feb 2026), not peer-reviewed as of June 2026; 11,871 bilingual EN/ZH MCQA items (Monolingual 7,013 / Comparative 4,310 / Parallel 548); CC BY 4.0. Author byline to be completed from the published version (repository CITATION.bib carries template placeholders).. doi:10.64898/2026.02.13.26346236
- [12] Sachiko Mita et al. 2026. A Comparison of Regional Decisions for Doses and Administrations of New Drugs: Concordance, Discordance, and Dependencies. *Clinical Pharmacology and Therapeutics* (2026). PMC13083379; 111 NMEs

- approved in the US and Japan 2017–2023; dosing concordance 77% PMDA–FDA, 84% EMA–FDA..
- [13] Antía Gómez-Fernández et al. 2026. Global Regulatory Variability in Small-Molecule Inhibitor Approvals: Differences in Timelines, Dosing, and Pediatric Indications across FDA, EMA, and PMDA. *Clinical and Translational Oncology* (2026). PMID 41045422; 55 small-molecule inhibitors approved across all three agencies..
 - [14] Daniele Sartori, Michele Fusaroli, Jeffrey K. Aronson, et al. 2026. PVverse/paper.PI: Multi-Country Product-Information Corpus (627 drug–ADR signals across 18 countries). <https://github.com/PVverse/paper.PI>. MIT-licensed R data package accompanying *Drug Safety* 2026; DoTS + SIM instruction-feature coding.. doi:10.1007/s40264-026-01666-6
 - [15] Yuqi Li, Junhao Dong, Jiao Liu, Piotr Koniusz, Hansheng Zeng, Chuanguang Yang, Junming Liu, Yingli Tian, Tingwen Huang, and Hao Wu. 2026. Evolving Multimodal Models for Physical Dynamics: A Multi-objective Neuroevolution Approach. *IEEE Transactions on Evolutionary Computation* (2026). doi:10.1109/TEVC.2026.3698641
 - [16] Guotao Liang, Zhangcheng Wang, Juncheng Hu, Haitao Zhou, Ziteng Xue, Jing Zhang, Dong Xu, and Qian Yu. 2026. Render-in-the-Loop: Vector Graphics Generation via Visual Self-Feedback. *arXiv preprint arXiv:2604.20730* (2026). doi:10.48550/arXiv.2604.20730
 - [17] Guotao Liang, Zhangcheng Wang, Chuang Wang, Juncheng Hu, Haitao Zhou, Junhua Liu, Jing Zhang, Dong Xu, and Qian Yu. 2026. VAnim: Rendering-Aware Sparse State Modeling for Structure-Preserving Vector Animation. *arXiv preprint arXiv:2605.01517* (2026). doi:10.48550/arXiv.2605.01517
 - [18] Shuang Zeng, Dekang Qi, Xinyuan Chang, Feng Xiong, Shichao Xie, Xiaolong Wu, Shiyi Liang, Mu Xu, and Xing Wei. 2025. JanusVLN: Decoupling Semantics and Spatiality with Dual Implicit Memory for Vision-Language Navigation. *arXiv preprint arXiv:2509.22548* (2025). doi:10.48550/arXiv.2509.22548
 - [19] Lan Hu, Yuting Xin, Binqi Shen, Hanyu Cai, and Lier Jin. 2026. CoDES: A Context-Efficient Framework for Enhancing Small Language Models via Domain-Specific Adaptation and Model Ensembling. *Preprints* (March 2026). doi:10.20944/preprints202603.1152.v1
 - [20] Yisong Chen, Chuqing Zhao, Yishan Zhong, Ziyu Wang, Jiazhaoh Shi, and Wenjia Zheng. 2026. Applying the Deep Learning–Sector–Governance (DLSG) Framework to the U.S. Healthcare System: Opportunities, Deployment Pathways, and Policy-Aligned Evaluation. *Journal of Technology Innovation and Society* 4 (3 2026), 1–18. doi:10.63646/j.jtis.2026.040101
 - [21] Sophie Vieujeau, Bruce E. Sands, Remo Panaccione, David T. Rubin, Vipul Jairath, Silvio Danese, Laurent Peyrin-Biroulet, Stefan Schreiber, Séverine Vermeire, Geert D’Haens, Axel Dignass, Parambir S. Dulai, Neeraj Narula, and Walter Reinisch. 2025. Comparison of the FDA and EMA Guidance on Drug Development in Ulcerative Colitis: An Expert Panel Review. *Journal of Crohn’s and Colitis* 19, 7 (2025), jjaf111. doi:10.1093/ecco-jcc/jjaf111
 - [22] Kim Boesen, Peter C. Göttsche, and John P. A. Ioannidis. 2021. EMA and FDA Psychiatric Drug Trial Guidelines: Assessment of Guideline Development and Trial Design Recommendations. *Epidemiology and Psychiatric Sciences* 30 (2021), e35. doi:10.1017/S2045796021000147
 - [23] Sally W. Schwarz and Clemens Decristoforo. 2019. US and EU Radiopharmaceutical Diagnostic and Therapeutic Non-clinical Study Requirements for Clinical Trials Authorizations and Marketing Authorizations. *EJNMMI Radiopharmacy and Chemistry* 4, 1 (2019), 9. doi:10.1186/s41181-019-0059-2
 - [24] International Council for Harmonisation. 2024. *ICH E11A Guideline on Pediatric Extrapolation*. Technical Report. FDA / EMA.
 - [25] J. Richard Landis and Gary G. Koch. 1977. The Measurement of Observer Agreement for Categorical Data. *Biometrics* 33, 1 (1977), 159–174.

A Prompts and Code

Full prompts (LLM judge system/user, Graph-RAG Stage 1 triplet-extraction, Stage 2 graph-classification), evaluation scripts, and dataset schema are available at the project repository: <https://github.com/chuchugo/cross-jurisdiction-regulatory-divergence>

LLM judge (`llm_judge.py`): system prompt defines AGREE/DIVERGE/SILENT with explicit absence framing; user prompt pairs `{fda_text}` and `{ema_text}`. Model: `claude-haiku-4-5-20251001`, temperature 1.0, max tokens 100.

Graph-RAG (`graph_rag.py`): Stage 1 extracts a policy node `{subject, obligation_level, requirement, conditions}`; Stage 2 classifies the aligned node pair. Hard rules: MANDATORY $\leftrightarrow$ PROHIBITED $\Rightarrow$ DIVERGE (no LLM call); both SILENT $\Rightarrow$ SILENT. Max tokens: 300.